

Lab 1: Requirements Elicitation & Modeling

System: ShopSphere E-Commerce System

1. ASR Documentation

The following table documents the 3 chosen Architecturally Significant Requirements (ASRs) and their impact on the system's architectural decisions.

ASR Type	Statement	Architectural Impact
High Scalability (NFR)	The system must be able to handle a surge of 10,000 concurrent active users during peak sales events.	This drives the choice towards distributed architectures (Microservices, Event-Driven) and requires stateless processing in key services (like Cart/Order lookup) and effective load balancing.
Strict Security (NFR)	The system must securely process payments and protect sensitive customer data without storing credit card information internally.	This drives the architectural decision to isolate the payment module from the core system. It requires using secure API gateways to communicate with external payment providers and implementing strong encryption standards.
High Availability (NFR)	The system must maintain a high uptime (e.g., 99.9%) to ensure continuous service for users, even during server failures.	This requires architectural choices such as redundant multi-server deployments, automated failover mechanisms, and database replication to eliminate any single point of failure.

2. UML Use Case Diagram

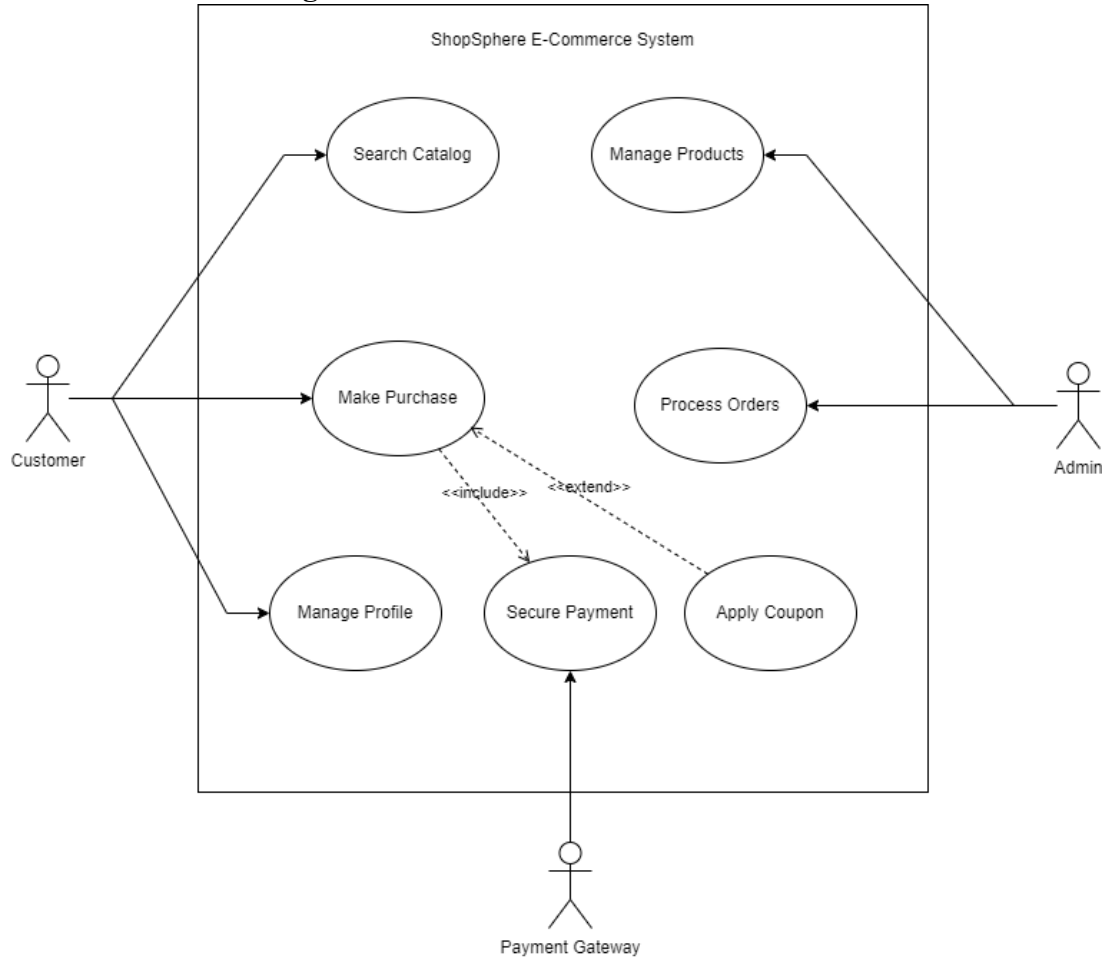


Diagram Checklist (Verify before submitting):

- **System Boundary:** A large rectangle labeled "ShopSphere E-Commerce System" representing the scope of the application.
- **Actors (Outside the boundary):** Web Customer, Administrator, and Payment Gateway.
- **Top-Level Use Cases (Inside the boundary):**
 - *Customer Use Cases:* Make Purchase, Manage Profile, Search Catalog.
 - *Admin Use Cases:* Manage Products, Process Orders.
- **Connections:** Solid lines connecting each Actor to their participating Use Cases.
- **Detailed Checkout Process (Critical):**
 - **<<include>> Relationship:** A dashed arrow pointing from *Make Purchase* to *Secure Payment*, indicating payment must happen as part of making a purchase.
 - **<<extend>> Relationship:** A dashed arrow pointing from *Apply Coupon* back to *Make Purchase*, indicating applying a coupon is optional during the purchase process.